

## DIEPENBEEK, Belgium

WMO: 064770

Lat: 50.9156N

Lon: 5.4503E

Elev: 39

StdP: 100.86

Time Zone: 1.00 (EUC)

Period: 09-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -6.5       | -4.4 | -11.7                         | 1.4 | -4.5 | -8.5 | 1.8 | -2.2 | 10.9                  | 8.7  | 9.7 | 9.7  | 2.2                   | 100  | 0.510 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.1                   | 30.9            | 20.3 | 28.2 | 18.7 | 26.2 | 17.7 | 21.1                | 28.3 | 19.9 | 26.3 | 19.0 | 24.4 | 2.8                  | 150  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.7                            | 13.6 | 23.2 | 17.8 | 12.8 | 22.2 | 16.9 | 12.1 | 21.4 | 61.0          | 27.9 | 57.1 | 26.1 | 54.1 | 24.6 | 25.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 8.7           | 7.5  | 6.6 | DB                         | -8.4 | 35.5               | 3.6 | 2.2                                                | -11.0 | 37.0       | -13.1 | 38.3       | -15.1 | 39.5       | -17.8 | 41.1 | (4) |
| (5)               |      |     | WB                         | -8.8 | 23.1               | 3.9 | 1.3                                                | -11.6 | 24.1       | -13.9 | 24.8       | -16.1 | 25.6       | -18.9 | 26.5 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 10.8        | 3.4   | 3.6   | 7.1   | 10.1  | 13.4  | 16.7  | 18.9  | 17.8  | 14.8  | 11.4  | 7.3   | 4.7   | (6)  |  |
|                                                                        | DBStd       | 6.41        | 4.07  | 4.18  | 3.29  | 3.67  | 3.69  | 3.06  | 3.30  | 2.74  | 2.84  | 3.65  | 3.65  | 4.11  | (7)  |  |
|                                                                        | HDD10.0     | 831         | 205   | 180   | 99    | 45    | 9     | 0     | 0     | 0     | 1     | 27    | 97    | 168   | (8)  |  |
|                                                                        | HDD18.3     | 2856        | 462   | 413   | 348   | 248   | 158   | 67    | 33    | 42    | 113   | 217   | 331   | 423   | (9)  |  |
|                                                                        | CDD10.0     | 1125        | 1     | 1     | 9     | 47    | 115   | 202   | 274   | 243   | 143   | 69    | 16    | 4     | (10) |  |
|                                                                        | CDD18.3     | 107         | 0     | 0     | 0     | 0     | 6     | 19    | 49    | 27    | 5     | 1     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 1279        | 0     | 0     | 0     | 16    | 82    | 223   | 551   | 317   | 82    | 8     | 0     | 0     | (12) |  |
|                                                                        | CDH26.7     | 394         | 0     | 0     | 0     | 1     | 12    | 58    | 198   | 107   | 18    | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 3.0         | 3.5   | 3.6   | 3.3   | 3.0   | 2.9   | 2.8   | 2.6   | 2.5   | 2.5   | 2.7   | 3.1   | 3.7   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 784         | 66    | 59    | 57    | 49    | 61    | 74    | 76    | 81    | 59    | 64    | 65    | 73    | (15) |  |
|                                                                        | PrecMax     | 1030        | 124   | 142   | 135   | 116   | 140   | 168   | 228   | 217   | 174   | 174   | 110   | 154   | (16) |  |
|                                                                        | PrecMin     | 600         | 3     | 5     | 2     | 4     | 22    | 14    | 31    | 16    | 15    | 30    | 6     | 19    | (17) |  |
|                                                                        | PrecStd     | 116         | 30    | 31    | 28    | 25    | 29    | 34    | 41    | 50    | 38    | 28    | 25    | 32    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 13.2  | 16.1  | 20.2  | 25.2  | 28.3  | 31.6  | 35.1  | 33.3  | 29.1  | 24.6  | 18.1  | 14.3  | (19) |  |
|                                                                        |             | MCWB        | 10.8  | 9.0   | 11.8  | 15.0  | 17.8  | 21.4  | 21.2  | 20.6  | 18.3  | 17.0  | 13.8  | 11.8  | (20) |  |
|                                                                        | 2%          | DB          | 11.9  | 12.0  | 18.1  | 23.0  | 25.7  | 28.3  | 31.6  | 29.9  | 25.7  | 21.3  | 16.1  | 12.9  | (21) |  |
|                                                                        |             | MCWB        | 9.9   | 8.8   | 10.6  | 13.6  | 16.5  | 19.0  | 20.1  | 20.5  | 17.5  | 16.4  | 13.1  | 10.8  | (22) |  |
|                                                                        | 5%          | DB          | 10.2  | 10.7  | 15.4  | 20.4  | 23.5  | 26.0  | 28.5  | 26.6  | 23.2  | 18.7  | 14.5  | 11.8  | (23) |  |
|                                                                        |             | MCWB        | 8.4   | 8.3   | 10.0  | 12.9  | 15.4  | 17.6  | 18.6  | 18.6  | 16.9  | 14.7  | 12.0  | 9.8   | (24) |  |
|                                                                        | 10%         | DB          | 9.1   | 9.6   | 13.1  | 17.5  | 21.2  | 23.6  | 26.3  | 24.3  | 20.9  | 17.1  | 13.1  | 10.3  | (25) |  |
|                                                                        |             | MCWB        | 7.5   | 7.5   | 8.9   | 11.8  | 14.5  | 16.7  | 17.8  | 17.5  | 15.9  | 13.9  | 10.9  | 8.7   | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 11.3  | 10.7  | 12.2  | 16.5  | 19.8  | 22.6  | 22.7  | 22.5  | 19.8  | 18.0  | 15.9  | 12.6  | (27) |  |
|                                                                        |             | MCDB        | 12.8  | 13.2  | 17.2  | 23.8  | 25.7  | 29.3  | 32.5  | 29.8  | 24.3  | 22.8  | 17.4  | 13.8  | (28) |  |
|                                                                        | 2%          | WB          | 10.0  | 9.5   | 11.4  | 14.5  | 17.9  | 20.3  | 21.0  | 20.7  | 18.3  | 16.5  | 13.4  | 11.3  | (29) |  |
|                                                                        |             | MCDB        | 11.7  | 11.4  | 16.2  | 21.2  | 22.5  | 26.5  | 29.0  | 27.2  | 23.0  | 20.7  | 15.4  | 12.6  | (30) |  |
|                                                                        | 5%          | WB          | 8.8   | 8.4   | 10.3  | 13.5  | 16.0  | 19.0  | 20.0  | 19.5  | 17.3  | 15.4  | 12.4  | 10.2  | (31) |  |
|                                                                        |             | MCDB        | 10.3  | 10.4  | 14.5  | 19.6  | 21.1  | 24.6  | 26.7  | 25.5  | 21.3  | 18.6  | 14.6  | 11.8  | (32) |  |
|                                                                        | 10%         | WB          | 7.6   | 7.3   | 9.3   | 12.5  | 14.8  | 17.7  | 19.0  | 18.6  | 16.2  | 14.5  | 11.2  | 9.0   | (33) |  |
|                                                                        |             | MCDB        | 9.0   | 9.4   | 12.5  | 17.6  | 20.2  | 21.9  | 24.2  | 23.2  | 19.7  | 16.8  | 13.4  | 10.5  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 5.0   | 6.3   | 9.2   | 11.0  | 10.6  | 10.3  | 11.1  | 11.0  | 10.7  | 8.4   | 6.3   | 5.2   | (35) |  |
|                                                                        |             | MCDBR       | 5.3   | 8.7   | 14.9  | 15.8  | 14.5  | 14.7  | 16.1  | 16.1  | 15.2  | 11.2  | 7.8   | 5.8   | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 4.5   | 6.2   | 9.0   | 8.2   | 6.9   | 6.7   | 6.5   | 7.8   | 8.5   | 6.8   | 5.6   | 4.2   | (37) |  |
|                                                                        |             | MCWBR       | 5.3   | 8.2   | 12.0  | 13.9  | 12.2  | 12.4  | 13.7  | 13.9  | 12.2  | 10.2  | 7.1   | 5.4   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.326 | 0.349 | 0.382 | 0.415 | 0.423 | 0.422 | 0.427 | 0.418 | 0.389 | 0.364 | 0.340 | 0.323 | (40) |  |
|                                                                        |             | taud        | 2.400 | 2.357 | 2.272 | 2.206 | 2.226 | 2.247 | 2.236 | 2.283 | 2.351 | 2.410 | 2.402 | 2.369 | (41) |  |
|                                                                        | Ebn at Noon |             | 695   | 773   | 809   | 823   | 833   | 836   | 825   | 812   | 795   | 745   | 667   | 630   | (42) |  |
|                                                                        |             | Edh at Noon | 63    | 84    | 109   | 130   | 133   | 131   | 131   | 119   | 100   | 79    | 62    | 56    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 0.77  | 1.44  | 2.60  | 4.14  | 4.94  | 5.37  | 5.08  | 4.33  | 3.24  | 1.91  | 0.91  | 0.60  | (44) |  |
|                                                                        | RadStd      |             | 0.10  | 0.28  | 0.37  | 0.47  | 0.46  | 0.47  | 0.61  | 0.43  | 0.33  | 0.16  | 0.10  | 0.09  | (45) |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (84 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page